FUSION-INTEGRATED · ASSIGNMENT 6B

Employee Leave Management

Phase 2 implementation and verification report

Prepared: 27 October 2026
Selected Phase-1 base: Implementation 1, end-of-6A commit `4a2fe133`, branch `feat/elm-module-integration`
Phase-2 commit: `0119b124` on `feat/elm-assignment-6b`

> **Current assessment: locally verified, not production-ready.**
> The application can authenticate a faculty test account and read ELM request history in the isolated preview. Real submission remains blocked by missing authoritative integration providers, and the existing PostgreSQL runtime schema has not been reconciled with the new ELM models.

120	35	33
ELM tests passed	frontend tests passed	files in Phase-2 commit

This report distinguishes verified behavior from unimplemented integrations and records the branch, tests, and remaining work for review.

1. Scope and implementation

Phase 2 continued from the selected P1-1 Fusion integration base. PR #238 added the initial ELM app, registry, API files, and host URL/settings wiring; the selected base is recorded at commit `4a2fe13`. The separate P1-2 PR #288 added a registry-only slice. Phase 2 expanded the P1-1 scaffold into a tested request-management slice.

Leave-day and request rules

- Inclusive calendar-day counting for EL, COL, and VL, including weekends and closed holidays.
- Same-day CL first-half/second-half requests charge exactly 0.5 day.
- Unsupported CL full-day/multi-day and RH/SCL quantities fail explicitly.
- Request validation covers category, dates, required fields, half-day metadata, overlap, and exact Decimal balance inputs.

Persistence and service boundaries

- Added ELM request, approval-step, substitute-assignment, and workflow-action models with migrations.
- Applicant identity comes from the authenticated principal; IAM identities remain external integer IDs.
- Submission validates before persistence and writes route/audit rows transactionally.
- Routing, balance, and VL calendar providers are explicit fail-closed adapter contracts.

Workflow and API

- Added recommendation/decision transitions, substitute response/replacement, withdrawal, and terminal-state guards.
- Added authenticated applicant-owned request history/detail, submission, decision, withdrawal, and substitute endpoints.
- History/detail queries are scoped to the authenticated employee; request input cannot supply applicant ID.

Fusion and frontend integration

- Registered the ELM module, navigation, permissions, import boundaries, and OpenAPI schema.
- Added the faculty request form and own-request history/detail UI, including pending withdrawal.
- Reviewer queues, substitute response UI, and print/export are outside this slice.

Change record

The submitted commit contains 33 files, 4,137 additions, and 30 deletions. It excludes an unrelated local edit to `devtools/management/commands/seed_demo.py`.

Scope limit: This is a first functional request-management slice, not completion of the full ELM baseline (32 CFRs / 128 AFRs). Balance accounting, calendar administration, hierarchy maintenance, year-end processing, SLA notifications, and other workflows remain open.

2. Verification evidence

Verification	Result	Evidence
`make check`	**Passed**	Django system and migration checks; Ruff; import-linter; module-boundary and plural-contract checks; navigation/route and permission-manifest checks; OpenAPI schema check; full backend pytest suite.
`.venv/bin/python -m pytest -o addopts='' modules/elm/tests -q`	**120 passed**	Focused deterministic ELM domain, API, submission, and workflow tests.
`make check-client`	**Passed**	TypeScript typecheck; Vitest (35 tests); production Vite build.
Local faculty session	**Passed**	Login 200; `/me` 200 with faculty ELM grant/navigation; request history 200; logout 200. Anonymous history returned 401.
Leave submission runtime	**Blocked, 503**	Submission returned `elm_dependency_unavailable` because the authoritative routing provider is not configured; request count remained zero.
PostgreSQL application schema	**Not verified for runtime**	The existing ELM table schema is incompatible with current models. The interactive preview used isolated SQLite; no shared-database migration was applied.

The project progress record also documents earlier focused ELM suite runs: 120 passed against the configured PostgreSQL test database and 120 passed with isolated SQLite test settings. These test-database results do not establish that the existing application PostgreSQL schema is ready.

Verification correction

The first full check stopped at OpenAPI generation because ELM views lacked schema annotations and schema discovery attempted an authenticated queryset. Added response/request schema annotations and a schema-only empty queryset guard, regenerated the OpenAPI file, then reran `make check` successfully.

Checks not completed

- `npm run lint` was unavailable because ESLint is not installed in the repository environment; no packages were added.
- No end-to-end submission or final-approval test against authoritative hierarchy, balance, and calendar services was possible.
- No independent security audit or production PostgreSQL migration was performed.

3. Remaining work and provenance

Known blockers

1. **Authoritative routing:** No effective-dated Unit Head/sanctioning hierarchy provider is configured. Valid production submissions therefore fail closed; ELM does not guess approvers from roles or demo data.
2. **Balance adapter:** No provider supplies an authoritative balance and idempotent final-approval revalidation/posting operation. Final approval cannot debit or reserve balances.
3. **VL calendar adapter:** No academic-calendar provider exposes the effective vacation periods needed for VL checks.
4. **Database reconciliation:** The old application PostgreSQL ELM table shape differs from the new model; migration application remains pending review. Local user-facing verification used SQLite.
5. **Broader ELM scope:** Admin policy/hierarchy maintenance, reviewer queues, substitute-response UI, print/export, leave-account events, resumption, year-end, notifications, SLA processing, and additional CFR workflows remain incomplete.

> **Deployment recommendation: do not deploy for operational leave submission yet.**
> Keep the build in a controlled test environment until provider contracts, PostgreSQL schema migration, and end-to-end integration tests are completed and reviewed.

Model and prompt provenance

The persistent project log `docs/elm-project/OLLAMA_PROMPT_HISTORY.md` records Qwen3.5:4b implementation and test drafts, retries, and corrections. The drafts were reviewed and corrected before verification. This report does not claim that every Phase-2 line was authored by that local model; the work also includes manual and Codex-assisted review/integration edits.

The Phase-1 source and current code are separate links: P1-1 PR #238; Phase-2 branch; Phase-2 commit 0119b12.

PR status: The branch is pushed to the Ishit-1290 fork. A PR to `FusionIIIT/Fusion-Integrated:main` has not been opened because this environment lacks authenticated GitHub API/CLI access. Open the comparison to create it. This comparison includes the unmerged P1-1 base commits.

Useful local evidence: `docs/elm-project/ELM_IMPLEMENTATION_PROGRESS.md`, `docs/elm-project/ELM_REQUIREMENTS_TRACEABILITY.md`, and `docs/elm-project/OLLAMA_PROMPT_HISTORY.md`. The prompt history is a local file, not a public URL; share the file separately with instructors.